

Проверка версии docker

```
docker --version
```

Создание рабочей директории

На диске "C" создаём папку "docker-system". В папке "docker-system" создаём файл "docker-compose.yml":

```
services:
  backend:
    image: mrdoka/network-admin-panel-backend:latest
    container_name: admin-panel-backend
    restart: unless-stopped
    depends_on:
      postgres:
        condition: service_healthy
      kafka:
        condition: service_started
    environment:
      SPRING_DATASOURCE_URL: jdbc:postgresql://postgres:5432/admin_panel
      SPRING_DATASOURCE_USERNAME: postgres
      SPRING_DATASOURCE_PASSWORD: postgres
      SPRING_KAFKA_BOOTSTRAP_SERVERS: kafka:9092
      KAFKA_TASKS_TOPIC: ${KAFKA_TASKS_TOPIC:-reconfig.tasks}
      APP_INSTANCE_ID: ${APP_INSTANCE_ID:-admin-panel}
      CORS_ALLOWED_ORIGINS: ${CORS_ALLOWED_ORIGINS:-http://localhost:3000,http://127.0.0.1:3000}
    ports:
      - "8080:8080"

  frontend:
    image: mrdoka/network-admin-panel-frontend:latest
    container_name: admin-panel-frontend
    restart: unless-stopped
    depends_on:
      - backend
    ports:
      - "3000:80"

  postgres:
    image: postgres:16-alpine
    container_name: admin-panel-postgres
    restart: unless-stopped
    environment:
      POSTGRES_USER: postgres
      POSTGRES_PASSWORD: postgres
      POSTGRES_DB: admin_panel
```

```
ports:
  - "5432:5432"
volumes:
  - postgres_data:/var/lib/postgresql/data
healthcheck:
  test: ["CMD-SHELL", "pg_isready -U postgres -d admin_panel"]
  interval: 5s
  timeout: 5s
  retries: 5

victoria-metrics:
  image: victoriametrics/victoria-metrics:v1.106.0
  container_name: admin-panel-victoria-metrics
  restart: unless-stopped
  command:
    - "-storageDataPath=/victoria-metrics-data"
    - "-httpListenAddr=:8428"
  ports:
    - "8428:8428"
  volumes:
    - victoria-metrics_data:/victoria-metrics-data

kafka:
  image: confluentinc/cp-kafka:7.6.0
  container_name: admin-panel-kafka
  restart: unless-stopped
  environment:
    KAFKA_PROCESS_ROLES: broker,controller
    KAFKA_NODE_ID: 1
    KAFKA_LISTENERS: PLAINTEXT://0.0.0.0:9092,CONTROLLER://0.0.0.0:9093
    KAFKA_LISTENER_SECURITY_PROTOCOL_MAP: PLAINTEXT:PLAINTEXT,CONTROLLER:PLAINTEXT
    KAFKA_ADVERTISED_LISTENERS: PLAINTEXT://kafka:9092
    KAFKA_INTER_BROKER_LISTENER_NAME: PLAINTEXT
    KAFKA_CONTROLLER_LISTENER_NAMES: CONTROLLER
    KAFKA_CONTROLLER_QUORUM_VOTERS: 1@kafka:9093
    KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR: 1
    KAFKA_AUTO_CREATE_TOPICS_ENABLE: "true"
    CLUSTER_ID: "MkU3OEVBNTcwNTJENDM2Qk"
  ports:
    - "9092:9092"

decision-engine:
  image: tem4ik4545/decision-engine:latest
  container_name: admin-panel-decision-engine
  restart: unless-stopped
  depends_on:
    postgres:
      condition: service_healthy
    kafka:
      condition: service_started
    victoria-metrics:
      condition: service_started
  environment:
```

```
KAFKA_BOOTSTRAP: kafka:9092
KAFKA_TASKS_TOPIC: reconfig.tasks
KAFKA_BOOTSTRAP_MARKER_ENABLED: "false"
POSTGRES_DSN: postgresql+asyncpg://postgres:postgres@postgres/admin_panel
VICTORIA_METRICS_URL: http://victoria-metrics:8428
METRICS_POLL_INTERVAL_SEC: "60"
DEDUP_WAITING_TIMEOUT_SEC: "600"
VLAN_RESOLUTION_METHOD: auto
ANOMALY_THRESHOLD: "3.0"
BANDWIDTH_THRESHOLD: "0.85"
QUARANTINE_VLAN: "999"
LINK_CAPACITY_MBPS: "100"
ICMP_THRESHOLD: "1000.0"
NEW_SOURCES_THRESHOLD: "10"
INTER_VLAN_RATIO_THRESHOLD: "0.4"
MAX_FLOW_BYTES_THRESHOLD: "1000000000"
SUBINTERFACE_PARENT: "GigabitEthernet0/0/1"
SUBINTERFACE_SUBNET_TEMPLATE: "192.168.{vlan_id}.1/24"
SUBINTERFACE_ENABLED: "true"
SEGMENTER_TYPE: greedy
LOG_LEVEL: DEBUG
```

netflow-collector:

```
image: scotix/avlan-netflow-collector:latest
container_name: avlan-netflow-collector
depends_on:
  - victoria-metrics
environment:
  VM_URL: http://victoria-metrics:8428
  NETFLOW_LISTEN_HOST: 0.0.0.0
  NETFLOW_LISTEN_PORT: "2055"
  TEMPLATE_TTL_SEC: "3600"
  LOG_LEVEL: DEBUG
ports:
  - "2055:2055/udp"
restart: unless-stopped
```

config-manager:

```
image: thmsshn/config_manager:latest
container_name: admin-panel-config-manager
restart: unless-stopped
depends_on:
  - kafka
  - postgres
environment:
  APP_NAME: config-manager
  APP_VERSION: 0.1.0
  DEBUG: "true"
  LOG_LEVEL: DEBUG
  DATABASE_URL: postgresql+psycpg2://postgres:postgres@postgres/admin_panel
  DB_ECHO: "false"
  DB_POOL_PRE_PING: "true"
  KAFKA_ENABLED: "true"
```

```
KAFKA_BOOTSTRAP_SERVERS: kafka:9092
KAFKA_REQUEST_TOPIC: reconfig.tasks
KAFKA_GROUP_ID: config-manager
CONNECTOR_TYPE: NETMIKO
QUEUE_ENABLED: "false"
QUEUE_URL: ""
APPROVAL_REQUIRED: "true"
AUTO_EXECUTE_TASKS: "true"
DEFAULT_ROLLBACK_ON_FAILURE: "true"
EXECUTION_TIMEOUT_SECONDS: "60"
MAX_PARALLEL_BATCHES: "4"
CHECKPOINTS_DIR: data/checkpoints
SSH_DEFAULT_USERNAME: admin
SSH_DEFAULT_PASSWORD: adminpass
SSH_ENABLE_SECRET: enablepass
SSH_DEVICE_TYPE: cisco_ios
SSH_CONNECT_TIMEOUT_SECONDS: "10"
SSH_GLOBAL_DELAY_FACTOR: "1"
SSH_SAVE_CONFIG: "false"
```

```
volumes:
  postgres_data:
  victoria-metrics_data:
```

После этого:

```
cd C:\docker-system
```

Скачиваем контейнеры:

```
docker pull tem4ik4545/decision-engine:latest
docker pull scotix/avlan-netflow-collector:latest
docker pull thmsshn/config_manager:latest
docker pull mrdoka/network-admin-panel-backend:latest
docker pull mrdoka/network-admin-panel-frontend:latest
docker pull postgres:16-alpine
docker pull victoriametrics/victoria-metrics:v1.106.0
docker pull confluentinc/cp-kafka:7.6.0
```

Запустить\остановить:

```
# Остановить контейнеры
docker-compose down

# Запустить контейнеры
docker-compose up -d

# Статус контейнеров
docker-compose ps
```

Проверка в браузере

```
# Админ-панель (frontend)
```

```
http://localhost:3000
```

```
login: admin
```

```
password: 1234
```

```
# Victoria Metrics
```

```
http://localhost:8428
```

```
# Админ-панель (backend)
```

```
http://localhost:8080/swagger-ui/index.html
```